Final Review — Guide to the Added Slides

Covers only the five new slides: Module 7, Module 8, Module 9, Module 10 and Final Implementation Status. Slides 1–14 are unchanged and are covered by the Review 2 guide.

What changed in the deck. Five slides were added after Slide 14, so the deck is now 21 slides. Four earlier slides were adjusted for consistency, because a final review cannot still say "60 %": Slide 9 now shows all ten modules filled in; Slide 10 was retitled "Modules 1 to 6"; Slide 13's caveat now points to Module 8 for test results; Slide 14 was retitled from "60 % Demo". The Literature Survey slides were verified byte-identical and were not touched.

The single most important change to understand. In Review 2 you said "the test split is held out and evaluated once." That has now happened — Module 8 reports test results. So the honest line is no longer "we are holding it back"; it is **"we held it back until every setting was fixed, then read it once."** Say it that way.

Slide 15 — Module 7: Fact Verification

What is on the slide

Top: the pipeline in one line. Left: retrieval quality, and the two label spaces. Right: the verifier results table and the key finding.

What the module actually does

Modules 1–6 answer "does this article *look* fake?" from its writing alone. Module 7 asks a different question: "is there *evidence* that supports or refutes this claim?" It works in two stages — first retrieve candidate evidence, then judge the claim against it.

Stage 1 — Retrieval, and why BM25

BM25 is a classical keyword-matching algorithm. It scores a document by how many of the claim's rare words appear in it. **Dense retrieval** (LaBSE) instead compares meaning-vectors.

Corpus	BM25 Recall@5	Dense Recall@5
FEVER (English)	0.5491	0.3227
X-FACT (Tamil)	0.6706	0.5347

Say: "We tested both retrievers and BM25 was clearly better on both corpora — 0.55 against 0.32 on FEVER and 0.67 against 0.53 on X-FACT. So BM25 is our primary retriever. That is a measured choice, not an assumption."

What Recall@5 means: out of the evidence that should have been found, what fraction appeared in the top five results. 0.67 means roughly two-thirds of the needed evidence was retrieved.

Stage 2 — The verifier, and why two label spaces

The verifier is an XLM-RoBERTa **cross-encoder**: it reads the claim and the evidence *together* and outputs a verdict. Two datasets are used, and their vocabularies are different:

- **FEVER** (English) — SUPPORTS / REFUTES / NOT ENOUGH INFO. This is textual entailment: does the evidence entail the claim?
- **X-FACT** (Tamil) — false / true / partly true or misleading. This is a fact-checker's verdict.

> **Never say "REFUTES" for a Tamil result.** The two label spaces mean different things. "Partly true or misleading" has no FEVER equivalent — forcing it into REFUTES would invent a judgement nobody made. Our code keeps them separate and the slide says so.

Reading the results table

Variant	Accuracy	Macro F1	What it tells you
FEVER claim-only	0.6066	0.6098	Verdict from the claim alone.
FEVER evidence-aware	0.6337	0.6356	Adding evidence: +2.6 points.
X-FACT claim-only	0.5792	0.2445	**Exactly the majority baseline.**
X-FACT evidence-aware	0.6380	0.4180	Adding evidence: **+17.4 points.**
X-FACT Tamil evidence	0.6244	0.4126	Restricting evidence to Tamil only.

> **Say — this is the strongest single result in the second half of the project:** "Look at the X-FACT claim-only row. Accuracy 0.5792, macro-F1 0.2445. Those are *exactly* the numbers a model gets by always predicting the majority class — it learned nothing from the claim text. Adding retrieved evidence raises macro-F1 by 17.4 percentage points. So on Tamil, essentially all of the verifier's skill comes from the evidence, not from the wording of the claim. Very few papers can demonstrate that this cleanly, because most never run the claim-only control on their own model."

Be ready for: "Why is 0.63 so much lower than your classifier's 0.97?"

They are different tasks. The classifier makes a **two-way** decision (fake or real) on news articles that come from systematically different sources. The verifier makes a **three-way** decision (supports / refutes / not enough info) about whether a specific claim is entailed by a specific piece of retrieved evidence — a much harder problem, and one where retrieval only finds the right evidence about two-thirds of the time. 0.63 on a three-class entailment task is a reasonable result; it is not comparable to a binary classification score.

Slide 16 — Module 8: Decision Fusion

What is on the slide

The fusion formula, the three frozen parameters, the test results table, and an honest box about how small the gain is.

What fusion means

We now have two opinions about every article: the classifier's P(fake), and the verifier's verdict. Fusion combines them into one number:

P(fake) fused = α × P(fake) classifier + (1 − α) × P(fake) verifier, then temperature scaling, then compare against the threshold.

The three parameters, and what each does

Parameter	Value	What it does
alpha (α)	0.95	How much to trust the classifier versus the verifier. 0.95 means the classifier supplies 95 % of the fused probability and the verifier 5 %.
threshold	0.56	The cut-off. At or above 0.56 the verdict is FAKE; below it, REAL. **Note it is not 0.5** — it was chosen to maximise validation macro-F1.
temperature	0.7079	Rescales the probability so it is better calibrated. Because it is below 1, it *sharpens* — pushes probabilities further from 0.5.

> **Why "frozen" matters.** All three numbers were fitted on the **validation** split, then locked. The test split was read once with those values fixed. If we had tuned them on test, the test score would be meaningless — it would measure how well we tuned, not how well the system generalises.

The results, and the honest reading

Macro F1	Classifier only	Integrated	Change
Overall	0.971786	0.972681	+0.000895
English	0.972654	0.973625	+0.000971
Tamil	0.961620	0.961620	0.000000

> **Say this before the reviewer works it out.** "Fusion improves overall macro-F1 by 0.000895 — that is under one thousandth. And for Tamil the two figures are identical to six decimal places: fusion changed no Tamil prediction at all on the test split. We report that as measured. The reason is structural: alpha is 0.95, so the verifier only contributes 5 % of the fused probability, and the verifier at 0.63 macro-F1 is much weaker than the classifier at 0.97. A small weight on a weaker signal produces a small effect."

If asked "then why include fusion at all?" — Because the alpha was *chosen by the data*, not assumed. We searched a grid of alpha values on validation and 0.95 won. That is itself a finding: it tells us that, with this verifier at this strength, the classifier should dominate. Had we forced a 50/50 blend, results would have got worse. Fusion also becomes the mechanism that lets a stronger future verifier contribute more, without redesigning the system.

Slide 17 — Module 9: Explainable AI

What is on the slide

Left: how occlusion works and how to read the sign. Right: a red box describing a defect the audit found, and how it was corrected.

What occlusion is

Take the article and record the model's P(fake). Then remove one word, run the model again, and see how much P(fake) moved. Repeat for every word. The change is that word's **attribution**.

Attribution	Removing the word...	Meaning
Positive	lowers P(fake)	that word was pushing the model toward FAKE
Negative	raises P(fake)	that word was pushing the model toward REAL

> **Say:** "Occlusion is simple and faithful: we do not model the model, we actually remove each word and measure what happens. 200 validation and 500 test explanations were generated, and verification passed 12 checks with zero failures."

> **Always add:** "These attributions explain the *classifier's probability*. They do not explain the verifier, and they do not prove anything is factually true or false."

The defect — how to present it (this is a strength, not a weakness)

An audit of the encoding path found a silent bug: the explanation module encoded text with **LaBSE**, while the classifier had been trained on **XLM-RoBERTa** mean-pooled, L2-normalised vectors. Both models output 768 numbers, so nothing crashed — the shapes matched. But the numbers meant completely different things.

Encoding path	P(fake) produced	Diagnosis
LaBSE	0.000000	right scale, wrong space
XLM-R without L2 normalisation	1.000000	right space, wrong scale
XLM-R with L2 normalisation	**0.0030934**	correct — stored value is 0.0030954

> **Say — practise this, it will impress:** "During Module 10 we ran a provenance audit on the encoding path and found a silent defect. The explanation module was encoding with LaBSE while the classifier expects XLM-RoBERTa embeddings. Both are 768-dimensional, so it produced no error — it just produced saturated probabilities, 0.000000 every time. We traced the exact code path that generated the stored embeddings, confirmed it uses XLM-RoBERTa with mean pooling and L2 normalisation, and verified the corrected path

reproduces a known stored value to within two parts in a million. Module 10 now refuses to start unless it can reproduce that value, so this class of bug cannot recur silently. We report the earlier explanation set as superseded rather than as a validated result."

Why this is a strength. A reviewer's job is to find weaknesses. If you present a defect you found yourself, diagnosed to the exact cause, fixed, and then guarded against with an automatic check — you have demonstrated engineering discipline that most student projects never show. Do not hide it and hope. Lead with it.

Slide 18 — Module 10: Application and Demo

What is on the slide

The end-to-end pipeline, what the interface shows, the encoder self-check with its measured values, the smoke-test result, and the testing summary.

What the application does

A user pastes one news article. The system runs the whole pipeline and displays: the verdict (REAL / FAKE / UNCERTAIN), the final P(fake), the confidence, the decision threshold, the detected language, the retrieved candidate evidence with sources and scores, the verifier's verdict in its own label space, the word-level explanation, and a deterministic reason.

The encoder self-check — the most important engineering feature

Check	Required	Measured
Cosine similarity against a stored embedding	≥ 0.999	0.9999999994
|P(fake) − stored P(fake)|	< 1e-4	2.056e-06

> **Say:** "Before the system will run at all, it re-encodes one record whose correct embedding and probability we already know, and checks it reproduces them. Cosine came out at 0.9999999994 and the probability differed by two parts in a million. If either check fails, inference refuses to start rather than producing a wrong answer quietly. That is the guard against the Module 9 defect."

> **Subtle point worth knowing.** Cosine similarity alone is not enough — it ignores scale, so an unnormalised vector still scores near 1.0. It was the *probability* check that caught the bug. Mention this if a reviewer asks why two checks are needed.

The smoke test — a real record, end to end

Output	Value
Detected language	English
Classifier P(fake)	0.003095
Verifier verdict	SUPPORTS (FEVER label space)
Fused P(fake)	0.002930
Confidence	0.994768

Decision threshold	0.56
Final verdict	**REAL**

> **Confidence is not P(fake) — do not conflate them.** P(fake) is 0.0029, meaning the model thinks the article is almost certainly real. Confidence is 0.9948, meaning the decision sits far from the 0.56 threshold. They are different quantities and the interface shows them separately. If asked: "Confidence measures how far the probability is from the decision boundary, expressed as a fraction of the available room. A P(fake) sitting exactly on the threshold would have confidence zero, no matter what the probability value was."

Testing summary

- 43 unit tests for the inference layer — all passed.
- 347 tests across the whole project — all passed.
- Module 10 output verification: 30 checks passed, 0 failed, 4 skipped (the skipped ones cover token attributions, which the smoke test ran without).

Slide 19 — Final Implementation Status

What to say (about 45 seconds)

> **Say:** "All ten modules are implemented and verified — 100 per cent of the proposed pipeline. The classifier reaches 0.9718 macro-F1 on the test split and the integrated system 0.9727. Every module has an independent verification pass: Module 7 with 23 checks, Module 8 with 12, Module 9 with 18, and Module 10 with 30 — all with zero failures."

> **Close on the honest note.** A good final sentence: "We have reported the results that flatter the project and the ones that do not — the fusion gain of under one thousandth, the Tamil figure that did not move, and the encoder defect we found and fixed. We would rather the work be correct than look impressive."

Likely questions on the new slides

Q1. Your verifier is only 0.63. Isn't that weak?

It is a three-class entailment task, not binary classification, and it depends on retrieval that finds the right evidence about two-thirds of the time. The meaningful comparison is against the claim-only control: on Tamil, claim-only scores exactly the majority baseline at 0.2445, and evidence raises it to 0.4180. The verifier is learning from evidence, which is what it was built to do.

Q2. Fusion barely helped. Was it worth building?

The alpha of 0.95 was selected by searching on validation, not assumed — so the small gain is itself the finding: with this verifier at this strength, the classifier should dominate. Building the fusion stage also means a stronger verifier can contribute more later without redesigning the system.

Q3. Why did the Tamil number not change at all under fusion?

Because alpha is 0.95 and the verifier's contribution on those records was never large enough to move any prediction across the 0.56 threshold. Identical to six decimal places means literally no Tamil prediction flipped. We report it rather than rounding it away.

Q4. If Module 9's explanations were wrong, why is Module 9 marked complete?

The module is complete in the sense that the attribution method, the pipeline and the verification are all implemented and passing. What the audit invalidated was the specific set of 700 explanations generated with the wrong encoder. The corrected encoding path exists and is guarded by the self-check in Module 10, so regenerating them is a re-run, not new development.

Q5. How do you know the test split was not used for tuning?

Module 8's verification includes a check that recomputes the test predictions from the stored parameter file. If any parameter had been changed after the test was read, that recomputation would disagree. It passed, along with 22 other checks.

Q6. Why is the threshold 0.56 and not 0.5?

Because 0.5 is only the natural cut-off if the two classes are equally frequent and equally costly. We searched thresholds on validation and 0.56 maximised macro-F1. It is shown in the interface so a user is never misled by a probability above 0.5 being reported as REAL.

Numbers to memorise for the new slides

Number	What it is
0.6706 vs 0.5347	BM25 vs dense Recall@5 on Tamil X-FACT — why BM25 was chosen.

0.5792 / 0.2445	X-FACT claim-only — exactly the majority-class baseline.
+17.4 points	Macro-F1 gained on Tamil by adding retrieved evidence.
α = 0.95, threshold 0.56, T = 0.7079	The three frozen fusion parameters.
0.971786 → 0.972681	Test macro-F1, classifier only vs integrated.
0.961620 both	Tamil test macro-F1 — fusion changed nothing.
0.9999999994 and 2.056e-06	Encoder self-check: cosine, and probability difference.
43 / 347 / 30	Inference tests, total tests, Module 10 verification checks.
10 of 10 = 100 %	Modules complete.